

Discovery Executive Summary

Project: discovery-14-aug · Generated: 14/08/2026, 11:54:38

EXECUTIVE SUMMARY

This report consolidates the overall ratings, key findings, and recommended actions from the 1 discovery analysis run across this codebase (frontend and backend). Each section below reproduces that analysis's executive view; full evidence and diagrams live in the individual reports.

Portfolio Overview

#	Analysis	Overall Rating
1	Testing & Quality Assurance Analysis	—

1. Testing & Quality Assurance Analysis

EXECUTIVE SUMMARY

The Klearcom platform has a critically thin test suite. The backend (Laravel 12 / PHP 8.3) ships with PHPUnit 11 but contains only 2 test files (3 test methods) — none of which exercise real business logic; they assert hardcoded arrays and random integers. All 6 API controllers, both service classes (RealTimeTestService, MongoService), the LegacyDataMapper, and all 4 Eloquent models are completely untested. The frontend (React 19 / TypeScript / Vite) has zero test infrastructure — no Jest, Vitest, Testing Library, Cypress, or Playwright is installed, and zero test files exist for the 13 source files including pages, components, hooks, and the API client. CI runs PHPUnit on every PR but with `coverage: none`, and the frontend CI job only runs `npm run build` — no tests are executed. Estimated overall test coverage is < 5% (backend) and 0% (frontend).

5.1 Benchmark Ratings Summary

#	Hotspot	Primary KPI	Good	Moderate	High Risk	Measured	Rating
H1	Untested Critical Logic	Critical modules with zero tests	0	1–3	>3	8 modules (6 controllers, 2 services)	High Risk
H2	Low Test Coverage	Overall coverage %	>80%	50–80%	<50%	~3% (estimated, backend ~5%, frontend 0%)	High Risk
H3	Missing Integration Tests	Boundaries covered %	>70%	30–70%	<30%	0% (zero integration tests)	High Risk
H4	Missing Contract Tests	APIs with contract tests %	>80%	40–80%	<40%	0% (0 of 16 endpoints)	High Risk
H5	Flaky / Skipped Tests	Skipped/flaky test count	0	1–5	>5	0	Good
H6	No CI Test Gate	Tests enforced in CI	Required gate	Runs, not required	No CI test run	Backend: runs, not required; Frontend: no test run	Moderate
H7	No Frontend Test Infrastructure (additional)	Frontend test framework installed + test files present	Framework + tests	Framework, no tests	No framework	No framework installed, 0 test files	High Risk
H8	Assertion-Free / Trivial Tests (additional)	Tests with no meaningful business assertions (target 0)	0	1–2	>2	3 (all existing test methods)	High Risk

5.4 Actions Required

Hotspot	Action	Rating	Priority
H1 — Untested Critical Logic	Write unit tests for RealTimeTestService, MongoService, LegacyDataMapper; write feature tests for all 6 API controllers	High Risk	Critical
H2 — Low Test Coverage	Enable coverage reporting in PHPUnit/CI; install Vitest for frontend; target 75% backend and 60% frontend coverage	High Risk	Critical

H3 — Missing Integration Tests	Add Laravel feature tests with RefreshDatabase; add MongoDB service to CI; test SSE streaming flow end-to-end	High Risk	Critical
H4 — Missing Contract Tests	Add assertJsonStructure tests for all 16 API endpoints; test validation rejection for mutation endpoints	High Risk	Critical
H6 — No CI Test Gate	Add frontend test step to CI; make both backend and frontend CI jobs required status checks on main	Moderate	Medium
H7 — No Frontend Test Infrastructure	Install Vitest + React Testing Library + JSDOM; write tests for useRealtimeTest hook, API client, pages, and store	High Risk	Critical
H8 — Assertion-Free / Trivial Tests	Replace all 3 trivial test methods with tests that exercise real application classes; remove tautological assertions	High Risk	High

5.5 Expected Outcomes

- **Critical business logic protected:** RealTimeTestService (IVR discovery, TFN reachability) and MongoService verified by unit tests, preventing silent regressions in the platform's core test execution and data pipeline.
- **API contracts enforced:** All 16 endpoints validated by contract tests, catching breaking response shape changes before they reach the frontend — especially the duplicated reachability calculation that could silently diverge.
- **Frontend safety net established:** Vitest + React Testing Library installed and wired into CI, covering the useRealtimeTest SSE hook, page components, and Zustand store — preventing UI regressions during refactors.
- **CI as a true quality gate:** Both backend and frontend test suites run on every PR with coverage reporting, and branch protection requires both jobs to pass before merge — no untested code reaches main.
- **Legacy code safely modernizable:** With tests around LegacyDataMapper's `extract()` patterns and LegacyReportController's unfiltered input, the upcoming tech-debt cleanup can proceed with confidence that behavior is preserved.
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